

AnyCam 3.1.0-rc1.0 Audit Report

Release audit - 04 October 2026 - live MJPEG and wide cards, stream refresh, card order - tag 3.1.0-rc1.0

Scope: phase 3.1.0 of the build plan (cards): B25, C19, D3, with CrystalHeeler's decisions of 2026-10-04: B25 automatic; C19 as proposed (MJPEG live, wide streams live on computers, still pictures on phones); D3 one order for every viewer. Built in the untested run CrystalHeeler ordered on 2026-10-04, on top of 3.0.1-rc1.0. This build is not field-tested.

1. Live MJPEG cards (C19)

New file `anycam_mjpeg.py`. A card for a camera with an HTTP MJPEG stream asks `/api/go2rtc/card` as before; the answer is kind 'mjpeg' and a WebSocket address. `go2rtc` is not involved, so this works while `go2rtc` starts.

Part	Design
One camera connection	<code>_MjpegHub</code> : one connection per camera, shared by every card that shows it; closed 5 s after the last card. Reconnects at 2 s, doubling to 30 s. The brand cooldown (Microseven) applies before each connection.
Login	Basic, then Digest on a 401. The Digest code moved out of <code>http_snap_loop</code> to <code>_http_digest_header</code> , unchanged.
Pictures	<code>_JpegSplitter</code> cuts JPEGs from the bytes by walking each JPEG's segments, so a wrong multipart boundary does not matter and an EXIF thumbnail's end marker does not cut a picture short.
To the page	One WebSocket message per JPEG. Not a long multipart HTTP response: the page builder notes that Home Assistant's ingress proxy ends those; the <code>go2rtc</code> WebSocket passes ingress without trouble.
In the page	The card shows the newest picture. A picture that arrives while one loads replaces the waiting one, so the card never falls behind. Each blob is freed once shown. A hidden page or an open Enhanced View closes the socket.
Stop	A refused password, a missing page or a single-picture URL stops the stream with a text message; the card shows still pictures and tries live again after 5 min.

2. Wide streams on computers (C19)

`_go2rtc_card_source(wide=True)` plays the smallest stream even when it is wider than 1,920. The page sends `wide=1` unless it runs on a phone (`navigator.userAgentData.mobile`, else the user agent). The Pi passes the bytes; the computer decodes them. The DVR sub-stream rule still comes first.

3. Saved streams read again (B25)

Part	Design
How	<code>_streams_refresh</code> runs <code>api_set_credentials</code> with the saved password, the same step as entering it by hand, which paces rate-limited cameras.
When	A card request that fails for a saved-stream reason (no stream small enough, no stream that can play live, an MJPEG or MPEG-4 stream, no URL); or 5 failed stream starts in the snapshot loop.

Part	Design
Limits	At most once every 6 hours for each camera. Not for a DVR channel card, an information card, or a camera with no saved password.
Failure	The saved streams stay; one warning line.

4. Card order (D3)

- A drag handle at the left of each card's name; pointer events, so a mouse and a finger both work. The card moves once, at the drop: moving a card restarts its live stream.
- The page posts the order of card keys (ip:port, #chN for a DVR channel) to `/api/card_order`. The add-on checks it (a list of up to 500 strings of up to 200 characters), drops duplicates and keeps it in `runtime.json`.
- `/api/cameras` returns the cameras in that order; cameras not in it follow in their own order. `renderGrid` puts the cards in that order, and moves nothing when they already are.

Tests run

Suite	Result	Scope
Release gate	PASS	Eight gates; 15 files; build inputs checked online.
Python integration	PASS	494 of 494; 23 new in section AB, including a real local camera server for the MJPEG relay.
JavaScript behaviour	PASS	132 of 132; new sections C19 and D3.
Undefined names	PASS	15 files, 0.
Not run	NOTED	No field test (CrystalHeeler's order for this run). The image was not built. Not checked: the MJPEG relay through Home Assistant's ingress, a real MJPEG camera, the drag on a touch screen, and a wide stream decoded by a real browser.

Best-practices compliance

Measured against *docs/AnyCam_Coding_Best_Practices.md*, Parts 1 to 6. Rows this change touches; the rest carry from the 3.0.1-rc1.0 audit.

Standard	Check applied	Result	Note
1.1 Truncated function	AST, contracts, undefined names	PASS	15 files.
1.4 Async	Tasks and waits	PASS	Hub and viewer tasks cancelled on close; no blocking call in the relay.
1.10 Exceptions	Network errors	PASS	Client errors caught at the connection; the card is told in words.
1.11 Resources	Connections, blobs	PASS	One camera connection per camera; closed after the last card; blobs freed.

Standard	Check applied	Result	Note
1.16 Refactoring	Digest helper	PASS	Moved to module level unchanged; http_snap_loop calls it.
1.18 Testing	New tests	PASS	23 server and 14 page checks.
2.x JavaScript	Async, events	PASS	Pointer capture; handlers removed at the drop; no await in forEach.
6.1 Security	Input at the boundary	PASS	/api/card_order validates type, count and length. The MJPEG socket serves only a camera that has an MJPEG stream; passwords never reach the page.
No import cycles	Imports	PASS	anycam_go2rtc takes _mjpeg_source and _streams_refresh through the main file's NEEDS.
Rule 9 (privacy)	New text	PASS	Tests use made-up addresses only.

Known issues carried forward

- An MJPEG stream over RTSP still shows still pictures (go2rtc has no MJPEG output loaded; adding it changes the go2rtc security model).
- Another viewer sees a new card order at the next page load.
- **B6:** the Microseven waits until CrystalHeeler unlocks it.
- **C21:** the classic engine stays as the fallback.

Acceptance (field test)

- Test system A: the Microseven card (HTTP MJPEG) plays live, about as fast as the camera sends; one camera connection with two browsers open.
- Test system A: the Hikvision PTZ card plays live on a computer; on a phone it shows pictures.
- After a camera's stream settings change, its card plays live again within one card retry (5 min) without entering the password.
- A card dragged to a new place stays there after a reload, and another browser shows the same order.

Files changed

File	Change
anycam_mjpeg.py	New: the MJPEG hub, the JPEG splitter, the WebSocket.
anycam_go2rtc.py	wide=1; the MJPEG answer; the B25 trigger.
anycam_credentials.py	_streams_refresh.

File	Change
anycam_snap.py	_http_digest_header; the B25 trigger at 5 failed starts.
camera_discovery.py	Card order; routes; imports; version 3.1.0-rc1.0.
page_script.py, anycam_page.py	MJPEG card, phone check, drag and order; CSS.
anycam_modules.py, Dockerfile	The new file.
tests/	Sections AB, C19, D3.
CHANGELOG.md, config.yaml, docs	3.1.0-rc1.0; module lists; build plan.